

Fiscal Consolidations in Commodity-Exporting Countries: A Small Open Economy DSGE Perspective*

Manuel González-Astudillo[†]

Juan Guerra-Salas[‡]

Avi Lipton[§]

April 22, 2026

Abstract

Fiscal consolidation in commodity-exporting economies is subject to substantial uncertainty because fiscal revenues depend heavily on volatile international commodity prices. This paper quantifies how deviations in commodity revenues affect consolidation outcomes using a dynamic stochastic general equilibrium model for a small open economy with a commodity sector, distortionary taxation, productive public spending, heterogeneous households, and external borrowing subject to an endogenous sovereign risk premium. The model is estimated using Ecuadorian data for 2004–2019 and then used to simulate Ecuador’s 2020–2025 IMF-supported fiscal consolidation under perfect foresight, treating the program’s fiscal instruments as fully anticipated policy paths. We find that historically plausible deviations in oil revenues generate pronounced asymmetries in fiscal and macroeconomic dynamics. Higher-than-expected revenues accelerate deficit reduction, lower sovereign spreads, and stimulate economic activity, potentially weakening incentives to sustain adjustment, while revenue shortfalls raise borrowing costs, depress investment, and intensify the required fiscal effort. These results highlight the importance of explicitly accounting for commodity-revenue uncertainty when evaluating consolidation programs. While the analysis is calibrated to Ecuador, the framework and qualitative findings are broadly applicable to commodity-exporting economies implementing fiscal adjustment under external financing constraints.

Keywords: Fiscal consolidation, commodity exporter, small open economy, DSGE model

JEL classification: E62, F34, F47, Q43

*The views expressed in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or the Central Bank of Chile.

[†]Board of Governors of the Federal Reserve System and Escuela Superior Politécnica del Litoral. E-mail address: manuel.p.gonzalez-astudillo@frb.gov.

[‡]Corresponding author. Central Bank of Chile. E-mail address: jguerra@bcentral.cl.

[§]Harvard University. E-mail address: alipton@g.harvard.edu.